just 0.2 sq mile (0.5 sq km) may contain more species of trees than the combined landmass of Europe and North America. Such biodiversity is vitally important to humans—not least because most new medicines are derived from plants, and so the eradication of the rainforest's rich store destroys potential cures for disease.

Rain forests, together with all other trees and woodland, also act like a sponge for rainfall. Tree roots drink up moisture and limit surface runoff. When forest is cut or burned, the soil is leached of many of its nutrients. If it covers a slope, the soil will wash away, leaving the land unfit for growing any kind of plants. Deep gullies may undermine trees that have not been cut, bringing them down. After heavy rains, catastrophic mudslides, which happen with increasing frequency, sweep down the slope, destroying everything in their path—including human settlements. In May 2014, for example, heavy rainfall on the deforested slopes of the Caribbean island of Hispaniola caused mudslides and floods that killed more than 2,000 people. Conversely, in extended periods of dry weather, exposed soil dries out faster than tree-covered areas, making it more prone to wind erosion.